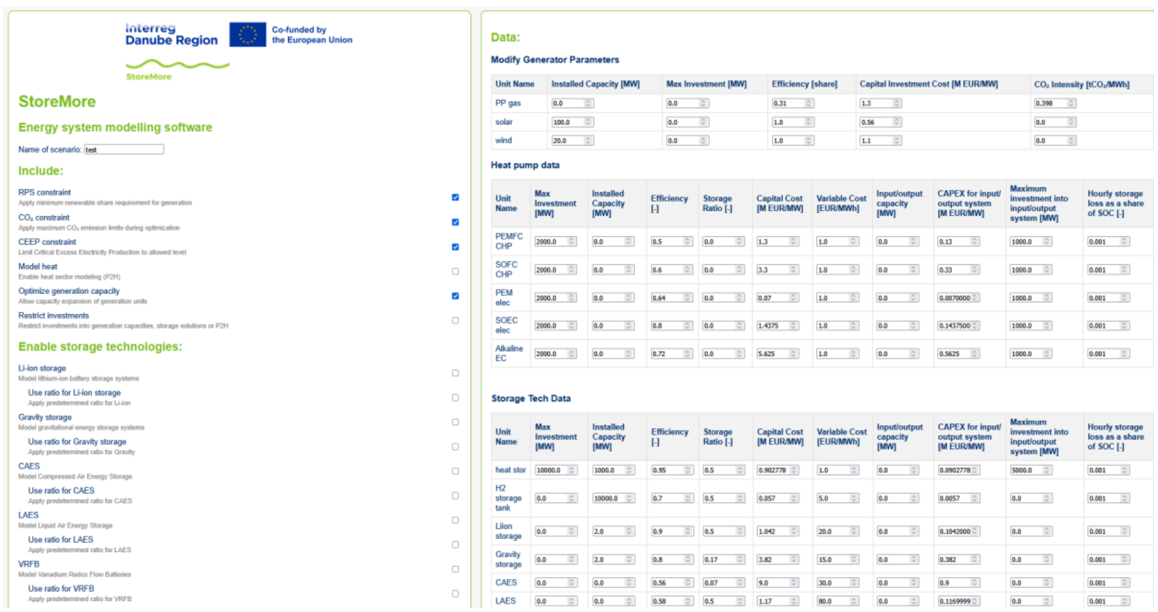
StoreMore optimization tool – User Manual

Luka Herc, University of Zagreb, Faculty of Mechanical Engineering and Naval Architecture

Version: August 2025

Project: Interreg Danube Region – Co-funded by the European Union



Sveučilište u Zagrebu
Fakultet strojarstva i brodogradnje

StoreMore

Table of Contents

Table of figures 3
1. Overview 4
2. Starting a New Simulation 5
  2.1 Entering Scenario Name 5
3. Constraints and Options 6
  3.1 Available Constraints 6
4. Storage Technologies 6
  4.1 Technology Selection 6
5. Input Data Configuration 7
  5.1 Generator Parameters 7
6. Technology-Specific Input Fields 8
  6.1 Hydrogen data 8
  6.2 Storage Technologies 9
  6.3. Heat pump data 10
  6.4. Thermal units 10
7. Model Parameters 11
8. Electricity Price Distribution Upload 12
9. Cost and Fuel Overview 13
  9.1 Annual Investment Costs 13
  9.2 Fuel Costs 14
10. Running the Simulation 15

Table of figures

Figure 1. Overlaps in the optimization stages 4
Figure 2. Divergence of the estimated from the real data 5
Figure 3. Entering the scenario name 5
Figure 4. Selection of the options to use 6
Figure 5. Selection of storage technologies....7
Figure 6. Data inputs for the genrators 8
Figure 7. Data inputs for the hydrogen technologies....9
Figure 8. Data inputs for the storage technologies 10
Figure 9. Data inputs for the heat pumps 10
Figure 10. Data inputs for the boilers 11
Figure 11. Parameters 12
Figure 12. Example of custom electricity price table....13
Figure 13. insertion of the custom electricity price table and the selection of country code 13
Figure 14. CAPEX data 14
Figure 15. Fuel cost data....15

1. Overview

StoreMore is a web-based tool for modeling, simulating, and optimizing future energy systems, focusing on electricity demand, generation, sector coupling, and storage technologies. It supports technology-specific constraints, investment planning, and energy balance evaluations.

Current configuration:

The model is run in two stages. The first stage includes the optimization of the entire planning horizon which is in this case the entire year. This stage gives the results on the necessary investments to optimize the system if the optimization function is enabled. Then, the results on the investments are saved and the second stage begins. This stage consists out of a range of 365 48 hour long partially overlapping optimization tasks. The planning horizon of each individual optimization step is divided into 2 parts. The first 24 hours use historical data from the tables for the distributions of electricity demand, renewable energy resource availability, electricity prices and heat demand. The second 24 hours use the “predicted” data that is based on the same distributions but has the uncertainties embedded. This replicates the uncertainties in the forecasts, and the system adapts to the slightly modified version of the reality. Then, in the next step, it is faced with the real data and has to adapt.

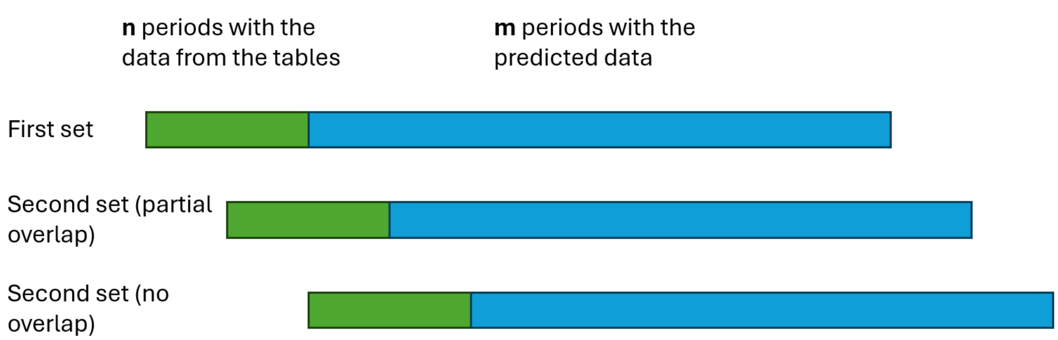


Figure 1. Overlaps in the optimization stages

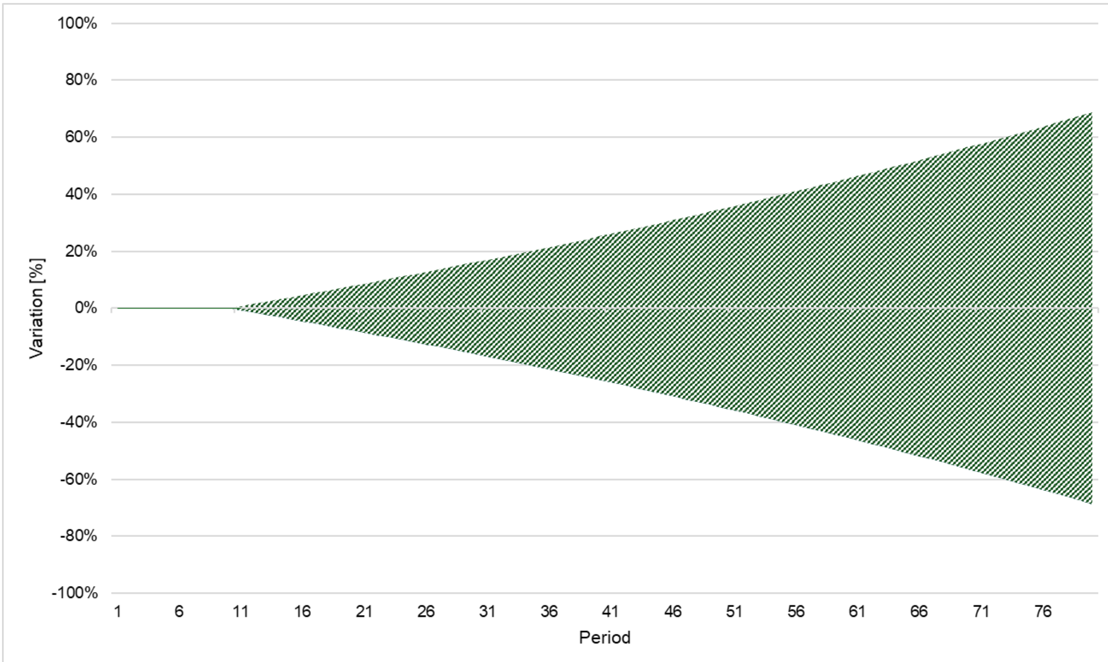


Figure 2. Divergence of the estimated from the real data

2. Starting a New Simulation

2.1 Entering Scenario Name

Field: Name of scenario

Action: Enter a unique name for the scenario (e.g., scenario1). This name is used for saving outputs and logs.

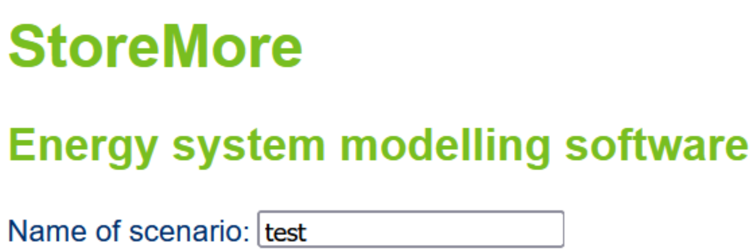


Figure 3. Entering the scenario name

3. Constraints and Options

3.1 Available Constraints

Enable any of the following by checking the corresponding box:

- RPS constraint – Minimum RES (Renewable Energy Source) share requirement.

- CO_2 constraint – Limit on total CO_2 emissions in the system.

- CEEP constraint – Restricts Critical Excess Electricity Production.

- Model heat – Enables sector coupling with heating. This will also activate the cells for inputting the data on heat pumps and boilers.

- Optimize generation capacity – Allows investment in new generation units.

- Restrict investments – Prevents expansion of generation, storage, or P2H capacity, etc. to the levels higher than the ones stated in maximum allowed investments.

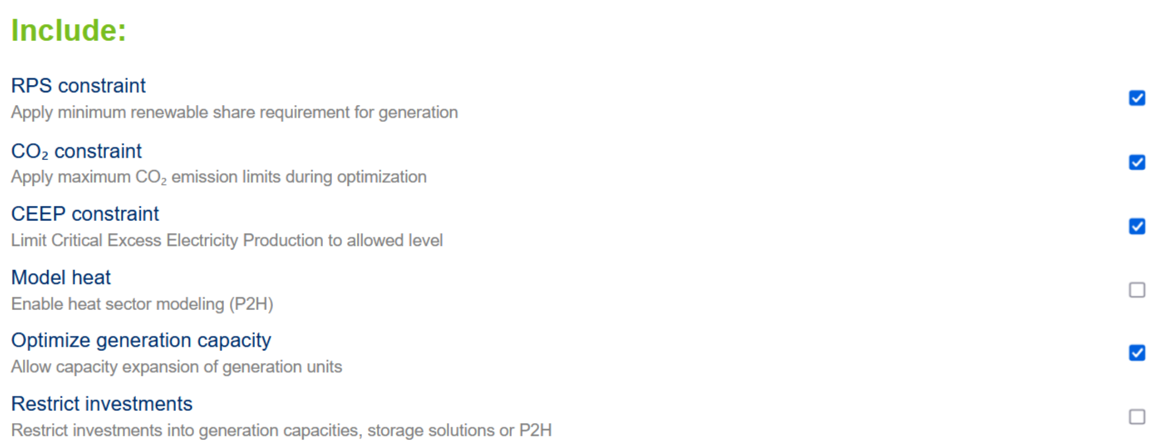


Figure 4. Selection of the options to use

4. Storage Technologies

4.1 Technology Selection

Check boxes to enable modeling of the following technologies:

- Li-ion storage

- Gravity storage

- CAES (Compressed Air Energy Storage)

- LAES (Liquid Air Energy Storage)

- VRFB (Vanadium Redox Flow Battery)

- SMES (Superconducting Magnetic Energy Storage)

You can also choose to use predefined ratios for each technology. If these are not selected, separate inputs on the charging/discharging capacity, CAPEX, OPEX need to be filled.

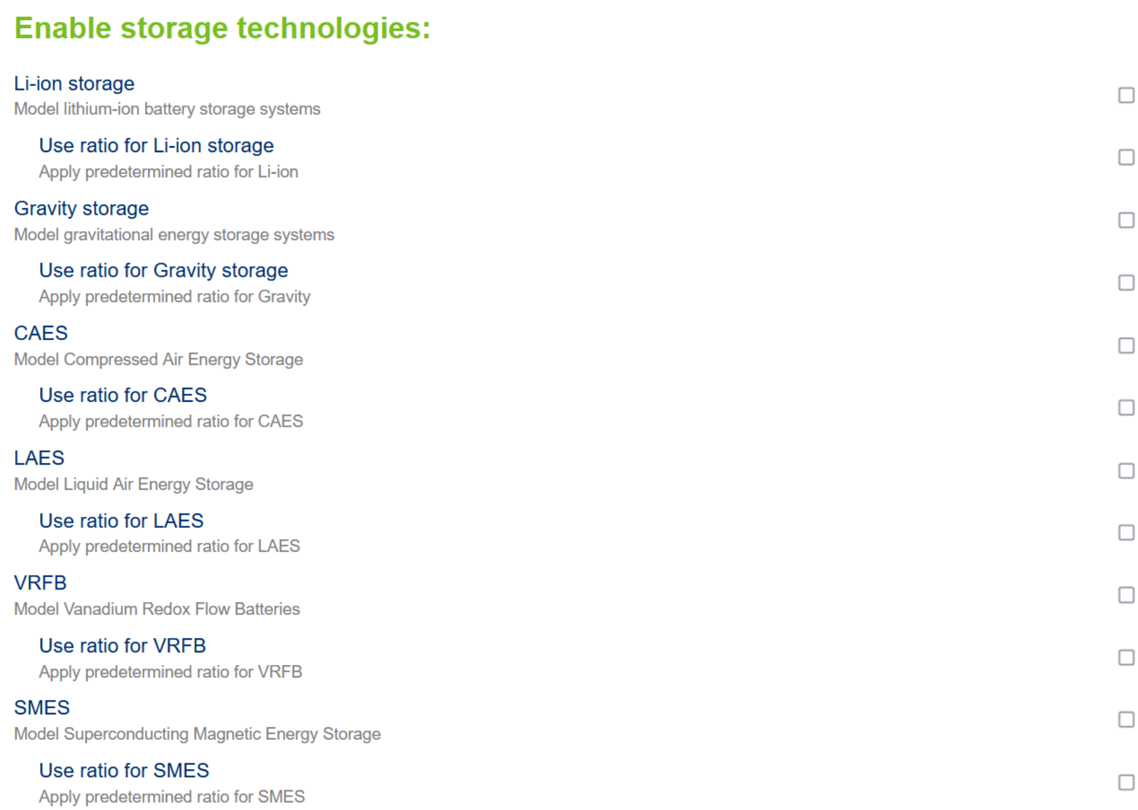


Figure 5. Selection of storage technologies

5. Input Data Configuration

5.1 Generator Parameters

Under 'Modify Generator Parameters', input values for:

- Installed Capacity [MW] – this is currently installed capacity ready to be used

- Max Investment [MW] – this is parameter that if the rule on restricting the investments is active will limit the investments or expansion of the capacities. The restriction is on maximum new capacity.

- Efficiency [0–1] – thermal efficiency of the power plant. In this case, this parameter is valid only for gas power plant as there is no fuel in wind and solar

- Capital Investment Cost [M EUR/MW] – equipment and installation cost of the 1 additional MW of capacity

- CO_2 Intensity [tCO_2/MWh] – specific emissions when generating electricity

Example generators: PP gas, Solar, Wind

Modify Generator Parameters

Unit Name	Installed Capacity [MW]	Max Investment [MW]	Efficiency [share]	Capital Investment Cost [M EUR/MW]	CO_2 Intensity [tCO_2/MWh]
PP gas	0.0	0.0	0.31	1.3	0.398
solar	100.0	0.0	1.0	0.56	0.0
wind	20.0	0.0	1.0	1.1	0.0

Figure 6. Data inputs for the genrators

6. Technology-Specific Input Fields

6.1 Hydrogen data

Configure PEMFC CHP, SOFC CHP, PEM elec, SOEC elec, Alkaline EC units with parameters like:

- Max Investment – the capacity that can be invested [MW]

- Installed Capacity – installed capacity of the technology [MW]

- Capital & Variable Costs

- Input/Output capacity, CAPEX

- Max investment into I/O system, Hourly SOC loss

H2 data

Unit Name	Max Investment [MW]	Installed Capacity [MW]	Efficiency [-]	Storage Ratio [-]	Capital Cost [M EUR/ MW]	Variable Cost [EUR/ MWh]	Input/output capacity [MW]	CAPEX for input/output system [M EUR/MW]	Maximum investment into input/ output system [MW]	Hourly storage loss as a share of SOC [-]
PEMFC CHP	2000.0	0.0	0.5	0.0	1.3	1.0	0.0	0.13	1000.0	0.001
SOFC CHP	2000.0	0.0	0.6	0.0	3.3	1.0	0.0	0.33	1000.0	0.001
PEM elec	2000.0	0.0	0.64	0.0	0.07	1.0	0.0	0.0070000	1000.0	0.001
SOEC elec	2000.0	0.0	0.8	0.0	1.4375	1.0	0.0	0.1437500	1000.0	0.001
Alkaline EC	2000.0	0.0	0.72	0.0	5.625	1.0	0.0	0.5625	1000.0	0.001

Figure 7. Data inputs for the hydrogen technologies

6.2 Storage Technologies

Configure storage units (e.g., heat stor, H2 storage tank, Li-ion, CAES, etc.) with:

- Max Investment – the capacity that can be invested [MW]

- Installed Capacity – installed capacity of the technology [MW]

- charging efficiency – share of the energy that ends up in storage [-]

- Storage Ratio – ratio between the storage capacity and the charging/discharging capacity

- Capital & Variable Costs

- Input/Output capacity, CAPEX

- Max investment into I/O system, Hourly SOC loss

Storage Tech Data

Unit Name	Max Investment [MW]	Installed Capacity [MW]	Efficiency [-]	Storage Ratio [-]	Capital Cost [M EUR/MW]	Variable Cost [EUR/MWh]	Input/output capacity [MW]	CAPEX for input/output system [M EUR/MW]	Maximum investment into input/output system [MW]	Hourly storage loss as a share of SOC [-]
heat stor	10000.0	1000.0	0.95	0.5	0.902778	1.0	0.0	0.0902778	5000.0	0.001
H2 storage tank	0.0	10000.0	0.7	0.5	0.057	5.0	0.0	0.0057	0.0	0.001
Liion storage	0.0	2.0	0.9	0.5	1.042	20.0	0.0	0.1042000	0.0	0.001
Gravity storage	0.0	2.0	0.8	0.17	3.82	15.0	0.0	0.382	0.0	0.001
CAES	0.0	0.0	0.56	0.07	9.0	30.0	0.0	0.9	0.0	0.001
LAES	0.0	0.0	0.58	0.5	1.17	80.0	0.0	0.1169999	0.0	0.001
VRFB	0.0	0.0	0.73	0.1	20.0	60.0	0.0	2.0	0.0	0.001
SMES	0.0	0.0	0.93	360.0	1.15	55.0	0.0	0.1149999	0.0	0.001

Figure 8. Data inputs for the storage technologies

6.3. Heat pump data

- Max Investment – the capacity that can be invested [MW]

- Installed Capacity – installed capacity of the technology [MW]

- variable cost – cost that incurs for each generated MWh of heat expressed in EUR/MWh

District Heating heat pumps Data

Unit Name	Max Investment [MW]	Installed Capacity [MW]	Variable Cost [EUR/MWh]
Air to water HP	0.0	0.0	0.0
Ground to water HP	0.0	0.0	0.0
Electric heater	0.0	0.0	0.0
Water to water HP	0.0	0.0	0.0

Figure 9. Data inputs for the heat pumps

6.4. Thermal units

- Max Investment – the capacity that can be invested [MW]

- Installed Capacity – installed capacity of the technology [MW]

- efficiency – thermal efficiency

- CO2 intensity – specific CO2 emissions for each MWh of heat generated

District Heating Thermal Units Data

Unit Name	Efficiency [-]	Installed Capacity [MW]	Max Investment [MW]	CO_2 Intensity [tCO2/MWh]
Gas	0.99	1000.0	0.0	0.2
Biomass	0.83	0.0	100.0	0.0
Oil	0.99	0.0	0.0	0.27
Coal	0.99	0.0	0.0	0.38
Geothermal	1.0	2.0	0.0	0.0

Figure 10. Data inputs for the boilers

7. Model Parameters

Enter or modify global model values:

- Total electricity demand [MWh]

- Total heat demand [MWh] (only visible if heating sector is active)

- Service life [years]

- No-load cost [€/MWh]

- No-load heat cost [€/MWh] (only visible if heating sector is active)

- Import/export capacity [MW] – this parameter defines how big the electricity exchange can be in each hour. If set to 0, model converts to island system with no influence of the external market.

- Import/export prices [EUR/MWh]

- Minimum RES share [%]

- Max CO_2 emissions [tCO_2]

- Max CEEP [% of demand]

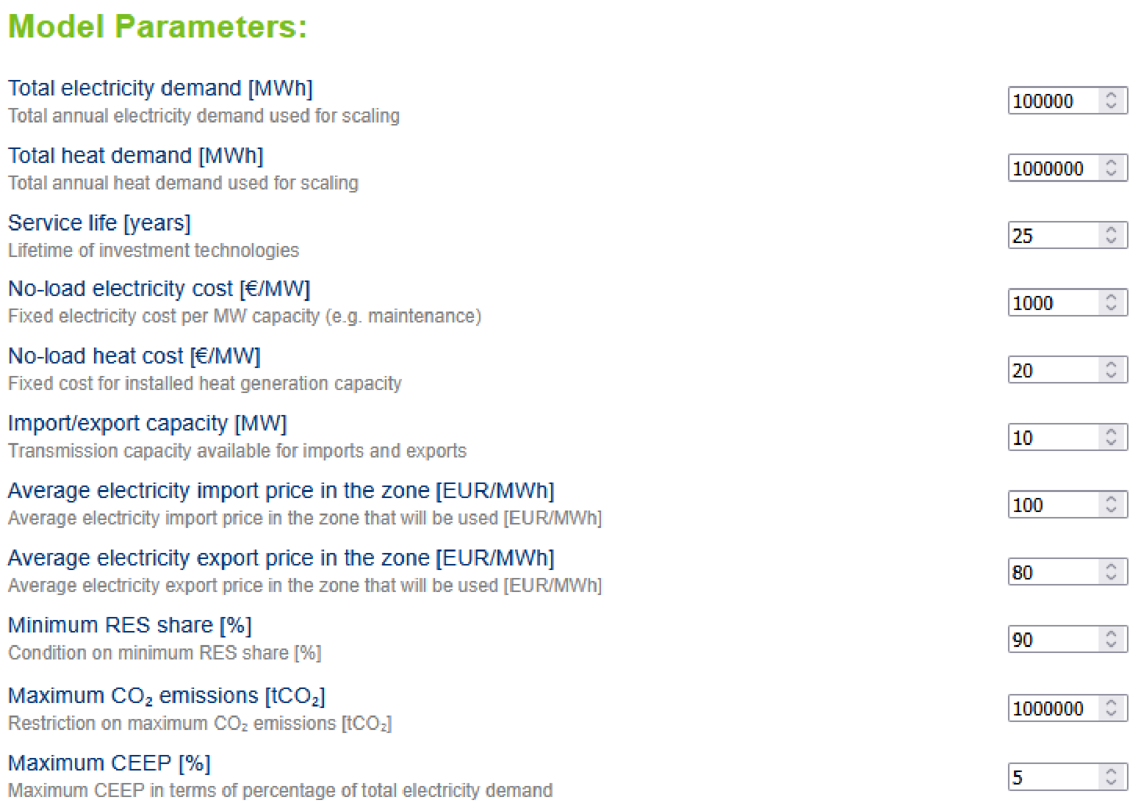


Figure 11. Parameters

8. Electricity Price Distribution Upload

At the bottom of the interface:

It is possible to upload custom electricity price distributions.

- Upload a CSV file with hourly electricity prices (optional). If this option is used, the file has to be in .csv format. The first column has to have the header “period” and the second column the header "<2 letter country code>” of the country you are using. Keep in mind that to get the final price of imports or exports that the model uses, the values in this table are multiplied with the average values described in section 7. Therefore, if the prices that are entered in the csv are intended to be used, use multiplication factors of 1 there.

period	HR
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1
9	1

Figure 12. Example of custom electricity price table

- Select Country code (e.g., HR, DE, FR) from the dropdown.

Figure 13. insertion of the custom electricity price table and the selection of country code

9. Cost and Fuel Overview

9.1 Annual Investment Costs

Displays costs for heat pumps, boilers, electrolyzers, and storage.

Annual Investment Cost

Technology	Investment Cost	Unit	Description
Air to water HP	0.86	M EUR/MW	Air-source heat pump for heating water
Ground to water HP	2.07	M EUR/MW	Ground-source heat pump for heating water
Water to water HP	0.48	M EUR/MW	Water-source heat pump for water heating
PEMFC CHP	1.3	M EUR/MW	PEM fuel cell combined heat and power unit
SOFC CHP	3.3	M EUR/MW	Solid oxide fuel cell combined heat and power
Electric heater	0.07	M EUR/MW	Standard resistive electric water heater
PEM elec	1.4375	M EUR/MW	PEM electrolyzer for hydrogen production
SOEC elec	5.625	M EUR/MW	Solid oxide electrolyzer for hydrogen production
Alkaline EC	0.902778	M EUR/MW	Alkaline electrolyzer for hydrogen production
Gas	0.278	M EUR/MW	Gas-fired boiler
Biomass	0.47	M EUR/MW	Biomass combustion boiler
Oil	0.278	M EUR/MW	Oil-fired boiler
Coal	0.47	M EUR/MW	Coal-fired boiler
waste	0.47	M EUR/MW	Waste-fired boiler
H2 storage tank	0.057	M EUR/MWh	Hydrogen storage tank (compressed or liquid)
Liion storage	1.042	M EUR/MWh	Lithium-ion battery storage
Gravity storage	3.82	M EUR/MWh	Gravity-based energy storage (e.g., lifting masses)
CAES	9.0	M EUR/MWh	Compressed Air Energy Storage
LAES	1.17	M EUR/MWh	Liquid Air Energy Storage
VRFB	20.0	M EUR/MWh	Vanadium Redox Flow Battery
SMES	1.15	M EUR/MWh	Superconducting Magnetic Energy Storage

Figure 14. CAPEX data

9.2 Fuel Costs

Editable prices in EUR/MWh for fuels: Biogas, Biomass, Coal, Oil, Gas, Diesel, etc.

Fuel Costs

Fuel	Cost	Unit
Biogas	7.16	EUR/MWh
Biomass	7.16	EUR/MWh
Coal	10.24	EUR/MWh
Nuclear	3.1	EUR/MWh
Oil	30.0	EUR/MWh
Diesel	130.0	EUR/MWh
Gas	60.0	EUR/MWh
Hydro	0.0	EUR/MWh
Solar	0.0	EUR/MWh
Wind	0.0	EUR/MWh
Hydrogen	1.0	EUR/MWh
Electricity	1.0	EUR/MWh
waste	3.58	EUR/MWh
Geothermal	1.0	EUR/MWh

Figure 15. Fuel cost data

10. Running the Simulation

Click Start simulation to run the model with current inputs.

Click Quit to exit without running.

The interface will display in which phase of the process the model is. In the second phase, when solving consecutive steps, the progress bar is visible.

After the model is complete, the link to download the results will appear. Only png and .svg files are downloaded.